

Luke Liang

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EDUCATION

BS in Computer Science, BS in Data Science, Math Minor (GPA: 3.86)

Aug 2021 - May 2024

Miami University

Oxford, OH

- Machine Learning (A), Predictive Models for Health (A), Object-Oriented Programming (A), High Performance Computing (A), Algorithms (A), Systems 2 (A+), Database Systems (A), Network Analysis and Modeling (A)
- Astronaut Scholarship (*national*), Choose Ohio First ML Scholarship, Summa Cum Laude, Honors College
- Received training at University of Tennessee (National Science Foundation sponsored) for ML and data science through the HPC at Oak Ridge national laboratory (*only student selected*).

PROFESSIONAL EXPERIENCE

Teaching Assistant (ML & Algorithms)

Aug 2022 - Dec 2023

Miami University

Oxford, OH

- Led and published a final research project that evaluated the performance of several ML models and data compression techniques (**Python, ML**).
- Tutored students in Python, data structures, algorithms, and ML.
- Provided feedback and graded exams, homeworks, quizzes, and projects for students.

Software Engineering Intern

Jun 2022 - Aug 2022

JPMorgan Chase & Co

Chicago, IL

- Converted existing AngularJS internal software to ReactJS with a full-time team. Deployed the updated framework that is used throughout the entire bank (**ReactJS, Jenkins**).
- Won the Code for Good Hackathon 2022 with a web-based project to revamp a non-profit organization's website with a group of other interns (**Javascript, MySQL**).
- Integrated legacy pages with new internally developed APIs and updated technical documents.

Research Assistant

Aug 2021 - October 2024

Miami University

Oxford, OH

- Published 4 papers focused on ML and network science (*1 first author, 3 co-author*) ([Google Scholar](#)). Two first author manuscripts involving web applications and simulation are under preparation.
- Created a self-contained web-based project via Docker that houses a simulation platform for researchers at the CDC to evaluate policy effectiveness on youth suicide prevention (**Python, Numba, Docker, HPC**).
- Applied ML models and several methods to augment input data to differentiate between erroneous and correct simulation models (**Python, ML, Tensorflow, NumPy, pandas**).
- Developed a simulation of HIV drug resistance and mutation to evaluate treatment plans (**Python NumPy**).
- Proposed a National Health Institute suicide modeling grant.

LEADERSHIP EXPERIENCE

Chapter President

May 2023 - May 2024

Tau Beta Pi - The Engineering Honor Society

Oxford, OH

- Established the chapter as an official club through student government to receive funding for recruitment, social, and service events.
- Increased membership and community engagement by over 70% after inactivity due to COVID.

PUBLICATIONS

Experimental evaluation of a machine learning approach to improve the reproducibility of network simulations

L. Liang, H. Phan, P. J. Giabbanelli

Simulation, 2024, ([link](#))

Simulation models for suicide prevention: a survey of the state-of-the-art

R. Schuerkamp, **L. Liang**, K.L. Rice, P. J. Giabbanelli

Computer 12.7, 2023, ([link](#))

Preserving simulation insight while removing data: verification of compressed simulation traces via machine learning

M. Nguyen, D. Vu, A. Vo, **L. Liang**, P. J. Giabbanelli

Annual Modeling and Simulation Conference (ANNSIM), 2023, ([link](#))

A new application of machine learning: detecting errors in network simulations

M. K. Wozniak, **L. Liang**, H. Phan, P. J. Giabbanelli

Winter Simulation Conference (WSC), 2022, ([link](#))

MANUSCRIPTS

A cellular automata approach to evaluating drug effectiveness for HIV

L. Liang, Z. Sun, P. J. Giabbanelli

In preparation

An artificial intelligence approach to support youth suicide prevention initiatives in the U.S.A

L. Liang, R. Schuerkamp, K. L. Rice, M. M. Brown, N. Nataraj, J. Mendoza-Alonzo, C. R. Harper, C. Florence, B. D.

Maldonado, B. Moran, P. J. Giabbanelli

In preparation

Impact of policies, practices, and programs on preventing adolescent suicide: an agent-based modeling approach

K. L. Rice, P. J. Giabbanelli, M. M. Brown, N. Nataraj, C. R. Harper, **L. Liang**, R. Schuerkamp, C. Florence

In preparation

PRESENTATIONS

Presenting author listed first

Oral Presentation. **L. Liang**, M. Nguyen, D. Vu, A. Vo, P. J. Giabbanelli, *Preserving simulation insight while*

removing data: verification of compressed simulation traces via machine learning, ANNSIM conference. May 2023

Oral Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli, *A new application of machine learning: detecting errors in network simulations*, WSC online. Dec 2022.

Poster Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli, *A new application of machine learning: detecting errors in network simulations*, COF Scholar showcase online. Feb 2022.